

IDAN MALKA

Embedded Software Engineer | Robotics & Control Systems

idanmalka7@gmail.com | +44 7508 857384 | LinkedIn | GitHub | Work permit: UK, EU, IL

PROFESSIONAL SUMMARY

Embedded systems engineer with 3 years building firmware, sensor drivers, and control systems for automotive, aerospace, robotics and energy sectors. I work in C/C++ on ARM Cortex-M (STM32) and Linux-based SBCs (Raspberry Pi) with FreeRTOS, and have hands-on experience with safety standards like MISRA-C and ISO 26262. From defining initial requirements to running validation tests on a proving ground, I enjoy every phase of project development.

TECHNICAL SKILLS

Languages: C (FreeRTOS, Bare-metal), C++, Python, Simulink, CAPL, MATLAB, Structured Text (IEC 61131-3)

Microcontrollers: ARM Cortex-M (STM32), ESP32, Raspberry Pi, Arduino

Protocols & Interfaces: CAN, MODBUS, ISO-TP (ISO 15765), RS485, SPI, UART

Standards: MISRA-C, MISRA-Simulink, ECSS, ISO 26262 (software & requirements traceability)

Testing & Debug: HIL testing, functional testing, unit testing, static analysis, oscilloscope, logic analyser

Tools: Git, Azure DevOps, Vector CANalyzer, Fusion 360, Excel, MongoDB

WORK EXPERIENCE

Embedded Software Controls Engineer | Proactsis Controls, UK Sep 2023 – Present

- Designed embedded software architecture and drivers for a satellite payload handling unit interfacing 8 communication channels (UART, RS485, CAN); validated via fault injection testing and resolved PCB faults using a logic analyser.
- Built a HIL test rig automating 800+ test cases/day on a simulated hypercar with twin front-axle motors.
- Implemented ISO 26262 anti-slip traction control, cutting slippage 90% on 25t EV; led wet-track testing.
- Developed a re-usable module to simultaneously control up to 24 high-performance Tesla 3-phase power converters for battery energy storage systems over CAN.

R&D Intern – Hardware Test Automation | Intuition Robotics, Tel Aviv, Israel Jun 2022 – Aug 2022

- Automated hardware QA for the ElliQ social robot, reducing test time during manufacturing.
- Built data acquisition tools to log and analyse sensor test data across 12 parameters (Python, batch).

KEY PROJECTS

UK Satellite Mission: Passenger Payload Integration | Project Lead, Proactsis Sep 2024 – Mar 2026

- Led firmware development in C/FreeRTOS for six UK pre-commercial experimental payloads.
- Integrated systems featuring AES-256 encrypted logging, ECSS-compliance, and fault detection/recovery.
- Developed a telemetry payload monitoring thermal robustness of a refuelling system over RS485.
- Featured in Space East Anglia journal for technology transfer from automotive to space.

Classic Vehicle Electrification: Multi-ECU Control & Diagnostics | Consulting Engineer Sep 2023 – Sep 2024

- Developed control and diagnostic strategies across VCU and BMS control units for classic-car EV.
- Implemented HVIL and isolation-resistance safety diagnostics, cell thermal-runaway detection, and crash-response logic; validated on-site with fault injection testing.

EDUCATION

MEng Mechatronic Engineering (First Class Honours) | University of Manchester, UK Sep 2019 – Jul 2023

- Built a modular robotic control system for radiation-cabinet chemical synthesis, boosting throughput 50% with remote control/monitoring at uL precision (best group project of the year).
- Led a team of 5 to build an STM32-based line-following robot integrating PID-based speed control.

Diploma of Higher Education – Advanced Maths & Science | Brussels, Belgium Jun 2019

ADDITIONAL

Languages: English (fluent), French (native), Hebrew (fluent)

Interests: Piano (15+ years, restaurant jazz pianist), clarinet, gym, volunteering sports coordinator at UoM